

REVIEW

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FinTech: a literature review of emerging financial technologies and applications

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Abstract

Future financing has the potential to evolve into a forward-looking and ingenious system that can enrich diverse technologies. This investigation focuses on five attractive emerging technologies tied to the finance, namely artificial intelligence (AI), machine learning, blockchain, augmented reality (AR)/virtual reality (VR), and quantum mechanics. Within the financial milieu, there is an ongoing pursuit for more precise, optimized, secure, and agile solutions capable of managing multifarious financial undertakings. Emerging technologies present remarkable modalities for achieving these objectives. This scholarly exposition expounds upon the creation of emerging technologies within the financial world, while concurrently deliberating upon the prospective applications of these technologies within financial systems. Given the nascent stage of development that characterizes these emerging technologies, the advancement of financial performance emerges as a substantial challenge owing to both technological and operational advantages. Technology-oriented financing has emerged as an essential trajectory for shaping the future of finance.

Keywords: FinTech, Emerging technology, AI (artificial intelligence), Machine learning, Blockchain, AR (augmented reality)/VR (virtual reality), Quantum mechanics

Introduction

This study investigates how emerging technologies are changing the field of finance and what the adoption of these technologies means for the future of the field. The use of technology in finance has grown significantly. An increasing number of cutting-edge technologies are emerging, creating a bright future for finance. In the early days, almost all financial products/ services were provided by financial institutions. Everything was bundled into one centralized location. Financial technologies, especially artificial intelligence (AI), machine learning, blockchain, augmented reality (AR)/virtual reality (VR), and quantum mechanics, provide a more diversified, convenient, and agile platform for the promotion of financial services and products, and play key roles in improving financial performance (Rundo et al. 2019; Zheng et al. 2019; Osmani et al. 2021).

FinTech is the application of technology to the delivery of financial services. However, FinTech did not appear suddenly in a vacuum. In 1918, the Federal Reserve Bank of the United States established a wire-transfer network using telegrams and codes. Previously, money movement between banks could only be achieved by delivering physical

cash or gold over long distances. In 1967, the first Automated Teller Machine (ATM) was installed at a branch of Barclays Bank in Enfield, North London. This technology allows customers to insert paper checks into ATMs for cash. This scenario is characterized by physical simulations of digital financial services. Not long after, during the 1970s, NASDAQ, the world's first digital stock exchange, was launched. This was followed by the development of a communication protocol between financial institutions to facilitate large volumes of cross-border payments. This trend continued into the 1980s with the rise of mainframe computers for banks. The growth of online banking has led to a shift in the way business is conducted and financial activities are perceived (O'Dair and Owen 2019; Israel et al. 2020; Cao et al. 2021).

Around 1998, PayPal was launched, giving a glimpse of what the new era of payments might look like as the world began to increasingly complete financial transactions online. This has led to FinTech thinking. In addition to attractive profits and competitive advantages, a lack of trust in institutions (i.e., banks and financial institutions, in relation to regulatory changes) has opened up the market to new FinTech-related entrants. Although this is a technological revolution in FinTech previous applications have been in the domain of established players (Zhang et al. 2020; Zhou et al. 2022; Aziz et al. 2022).

A new wave of innovation, driven mainly by new players and start-ups, has given rise to FinTech recognition. The iPhone ushered in the global adoption of smartphones, and mobile devices have become the primary way in which people access the Internet and their financial services. The Blockchain and cryptocurrencies have become mainstream. Mobile and P2P payments, digital lending, and cross-border mobile wallet remittances are examples of FinTech driven innovation activities. The rise of financial technology is based on decentralization, disintermediation, and intelligence, and banks are trying to occupy the entire financial value chain. FinTech focuses on providing financial products and services, focusing on the best user and service experiences (Ghoddusi et al. 2019; Cao 2022; Song et al. 2023).

The exploration of emerging financial technologies (AI, machine learning, blockchain, AR/VR, and quantum mechanics) and their applications in future finance stems from the growing recognition of the transformative impact of technology on traditional financial systems. Understanding the potential benefits and challenges of these innovations is critical to avoid industry changes and foster informed decision-making in the financial world. With the development of emerging technologies, an increasing number of financial services use cases are apparent. Fintech is expected to disrupt the traditional financial services industry by creating value chains and setting new standards for financial facilitation in the business world. This draws consumers to the new digital services and attracts their attention. The ultimate requirements for FinTech success are value creation and a customer-centric orientation. Success involves identifying the needs of all customers and finding a way to satisfy them holistically with the least friction. To achieve this, we must leverage the implementation of emerging technologies to effectively learn and adapt to the changing needs of customers. FinTech must keep pace with the times and to continuously develop advanced solutions (Cai et al. 2019; Goodell et al. 2021; Gan et al. 2021).

Through a review and summary of relevant literature, our objective is to explore the integration of emerging technologies into the financial system. We aim to understand

how these technologies address existing challenges, identify potential applications/scenarios for their adoption in the financial system, anticipate the challenges that may arise from their implementation, and analyze the potential transformative impact of technology-oriented finance on the future of financial systems and the broader financial landscape. Emerging technologies serve as a complementary mechanism for modernizing the financial system, making it more intelligent, agile, secure, problem-recognizing, and conducive to informed decision-making.

This study provides a relatively comprehensive review aimed at illustrating the integration of emerging technologies into the financial ecosystem, a dimension that previous studies have overlooked. The paper is structured as follows (Fig. 1): Sect. describes the "Methodology and data" procedures. Section interprets "The integration

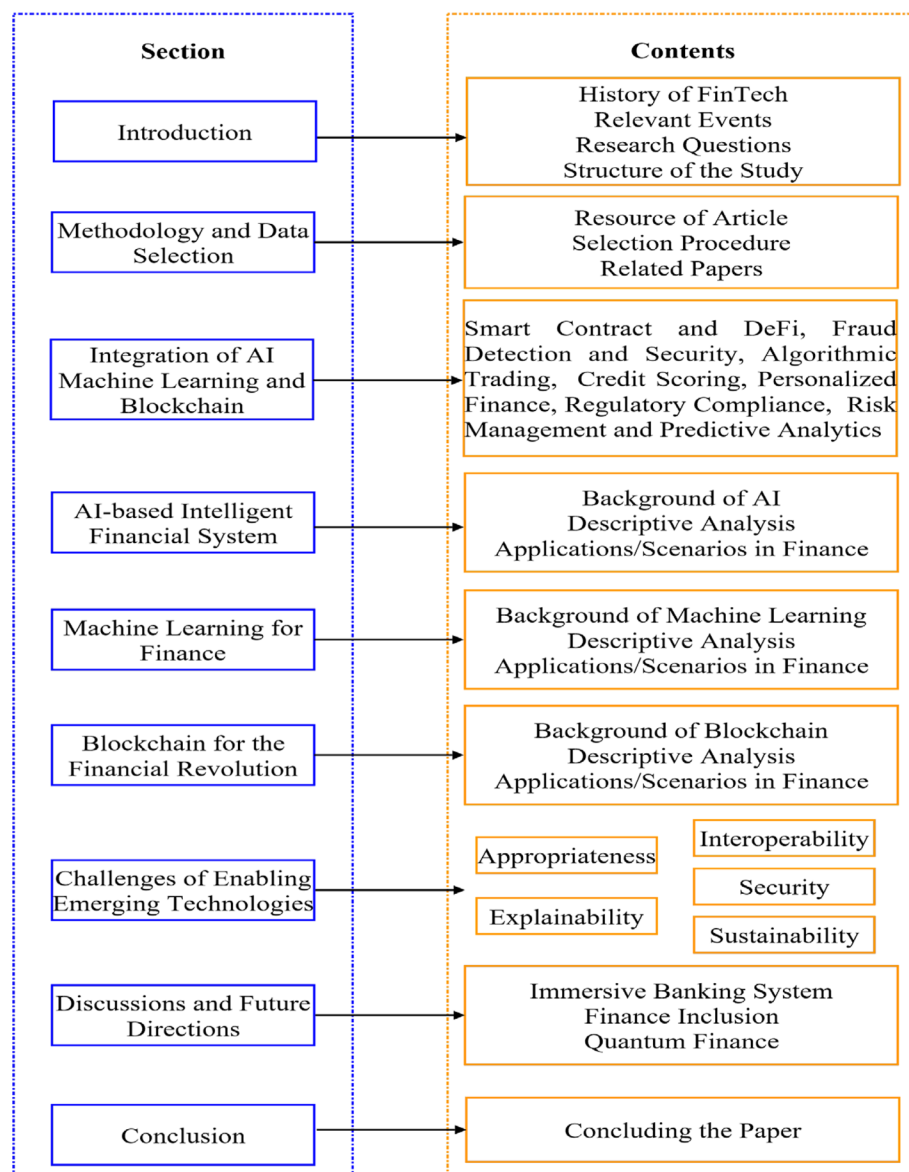


Fig. 1 Structure of the study

of AI, machine learning, and blockchain technology into finance”. Section presents extant studies on "AI-based intelligent financial systems". Section depicts an overview of "Machine learning for finance". Section describes "Blockchains in the financial revolution". Section addresses the "Challenges in enabling emerging technologies in finance". Covers "Discusses future research directions". Finally, sect. "Concludes" the paper.

Methodology and data

This study examines emerging technologies in finance. Specifically, AI, machine learning, and blockchain are the three core topics discussed, whereas AR/VR and quantum mechanics are discussed in the Future Directions section. Machine learning is a subset of AI; however, much of the research is directly related to financing based on machine learning, such as learning algorithms and mathematical models. Thus, we dedicate a separate section to machine-learning-based finance. Furthermore, these two emerging technologies are currently experimental, particularly in quantum mechanics, and a stable quantum computer is not yet available. This study provides cutting-edge explanations and insights into enabling emerging technologies in the financial sector.

All selected articles were obtained from the WoS (Web of Science), Scopus, and EBSCO databases. Because we are trying to depict a relatively comprehensive picture of emerging technologies in finance, more technologies are included in this article. Judging from the literature, there are many papers on emerging technologies in finance. We used keywords, " AI," "Machine Learning," "Blockchain," "AR/VR," "Quantum Mechanics," etc., plus "Finance" or "Financial Systems" to search for articles by title. We then checked all the searched articles for suitability. Specifically, we found that artificial intelligence (50), machine learning (96), and blockchain-based (150) finance-related research were the main studies (Table 1). However, for AR/VR and quantum mechanics, owing to their developmental stages, fewer papers were found that passed our search for suitability. This is another reason AR/VR and quantum-based financing will be considered in the future.

Among the selected top journals, many published papers (39) on Financial Innovation on how emerging technologies, such as AI, machine learning, and blockchain, transform and innovate finance and accounting systems, which have the potential to deeply impact and significantly change foundational infrastructures and operational modes of business ecosystems.

Table 1 Number of papers related to each topic

Emerging technology	Number of articles
AI	50
Machine learning	96
Blockchain	150
AR/VR and quantum mechanics	A Few
Total	296+

The integration of AI, machine learning and blockchain in finance

The confluence of AI, machine learning, and blockchain in the financial domain has the potential to redefine the provisioning of financial services, enhance security protocols, optimize operational efficacy, and expedite paradigms governing decision-making. The ensuing developments elucidate the mechanisms by which the integration of these technological facets engenders a transformation in the contours of financial discipline (Davenport and Ronanki 2018; Xu et al. 2019; Ali et al. 2020; Dwivedi et al. 2021; Van et al. 2021; Martin and Nagel 2022).

Smart contracts and decentralized finance (DeFi)

Blockchain technology is setting up a transformative era for the foundational underpinnings of Smart Contracts and Decentralized Finance (DeFi). Smart contracts, which denote programmatically executed agreements, fundamentally reconfigure contractual interactions by virtue of their autonomous execution characteristics. DeFi platforms, harnessed by incorporating these contracts into blockchain frameworks, are effectuating a paradigm shift in financing. These platforms provide a diverse spectrum of services, encompassing lending, borrowing, trading, and yield farming, all orchestrated seamlessly and devoid of intermediaries. The synergistic unity of AI and machine learning augments these strides, fostering intricate data analysis and risk-appraisal capabilities to augment the performance of DeFi services, thereby elevating operational efficiency and reliability to unprecedented levels (Cai 2018; Cong and He 2019; Chiu and Koepl 2019; Cong et al. 2022; Chod and Lyandres 2023).

Fraud detection and security

The utilization of AI and machine learning algorithms considerably fortified the domain of fraud detection and security in financial transactions. Through sophisticated scrutiny of patterns, deviations, and user behavioral dynamics, these algorithms function expeditiously and in real time, facilitating the prompt recognition of potentially illicit endeavors. This proactive modality, beyond deterring fraud, concurrently functions as a protective bulwark for client assets, aiming to build an environment of enhanced financial security (Manela and Moreira 2017; Van et al. 2017; Baker et al. 2020; Sokolov 2021; Pagnotta 2022; Wei et al. 2022; Akey et al. 2022).

Algorithmic trading

Algorithmic trading benefits from the integration of AI and machine learning. These technologies can be used to analyze financial datasets, identify patterns, and make rapid trading decisions based on predefined rules. Integrating the blockchain into this process can enhance transparency and auditability, making it easier to track and verify trad execution (Mestel et al. 2018; Weller 2018; Gómez Martínez et al. 2019; Gu et al. 2020; Leippold et al. 2022; Brogaard and Zareei 2023; Van et al. 2023).

Credit scoring and underwriting

Machine learning algorithms enhances credit risk assessment by analyzing extensive datasets consisting of transaction records, social media interactions, and other

data-related elements. Additionally, blockchain-powered identity verification adds a layer of trustworthiness and security to credit scoring procedures and collectively fosters more dependable and resilient underwriting practices (Zhu et al. 2019; Zhang and Ram 2020; Braggion et al. 2020; Gao et al. 2023).

Personalized financial services

AI can dissect customer behaviors and preferences such as customized investment guidance, personalized insurance schemes, and finely tuned budgeting suggestions. The incorporation of a blockchain, with secure and decentralized attributes, ensures the safeguarding of sensitive user information (Gomber et al. 2018; Easley et al. 2019; Kumar et al. 2019; Huang et al. 2019; Du and Xie 2021).

Regulatory compliance and auditability

The inherent immutability of blockchain safeguards financial transactions and records against tampering, to ensure a transparent and easily auditable trail. The synergy between AI and machine learning facilitates the automation of compliance procedures, helping financial institutions remain aligned with regulatory guidelines and industry standards (Yu et al. 2018; Frattaroli 2020; Han et al. 2023; He et al. 2023; Ferreira et al. 2023).

Risk management and predictive analytics

The integration of blockchain, AI, and machine learning led to more robust risk management strategies. These technologies analyze historical data, market trends, and external variables to predict potential risks, thus enabling proactive decision-making and optimization of risk management practices (Adamopoulos et al. 2021; Evgeniou et al. 2022; Dong et al. 2022a, b; Fuster et al. 2022; Obaid and Pukthuanthong 2022).

Overall, the integration of blockchain, AI, and machine learning in finance can drive innovation, improve efficiency, reduce costs, and enhance customer experiences while ensuring a secure and transparent financial ecosystem. These emerging technologies (AI, machine learning, blockchain, AR/VR, and quantum mechanics) are contributing to the development of more precise, optimized, secure, and agile financial solutions (Table 2). However, challenges and issues related to scalability, data privacy, and regulatory compliance have been addressed to fully unlock the potential of these technologies in the financial industry.

AI-based intelligent financial system

Emerging technologies, including AI, have revolutionized the world in many ways. The term “artificial intelligence” was first introduced in 1956, by Dr. John McCarty at a Dartmouth conference. He defined AI as “the science and engineering of making intelligent machines. AI is a technology that makes machines work and behave like humans. More recently, AI has accomplished this by creating machines that are widely used in health-care, robotics, marketing, business analytics, and many other applications. AI has already entered daily life such that most people are unaware that it is being used. It involves the development of computer systems that can perform tasks that typically require human intelligence. This includes tasks such as visual perception, speech recognition, decision

Table 2 Emerging technologies contributing to the development of more precise, optimized, secure, and agile financial solutions

Artificial intelligence (AI)

Precision AI enables precise data analysis, helping financial institutions extract valuable insights from datasets. It enhances accuracy in risk assessment, fraud detection, and customer profiling

Optimization AI-driven algorithms optimize processes such as portfolio management and trading, adapting strategies based on real-time market conditions

Security AI enhances security through anomaly detection, recognizing patterns indicative of fraudulent activities and ensuring swift responses

Agility AI's adaptability allows financial systems to quickly respond to changing market dynamics, regulatory requirements, and customer expectations

Machine learning (ML)

Precision ML algorithms improve precision in financial forecasting, credit scoring, and investment decisions by learning from historical data patterns

Optimization ML contributes to portfolio optimization, ensuring investment strategies align with risk tolerance and market conditions

Security ML aids in the detection of unusual patterns and behaviors, enhancing cybersecurity measures and protecting financial assets

Agility ML's ability to learn and adapt in real-time fosters agility, enabling financial institutions to respond swiftly to market shifts

Blockchain

Precision Blockchain ensures precision by providing a transparent and immutable ledger for transactions, reducing errors and ensuring accurate record-keeping

Optimization Smart contracts on blockchain automate and optimize various financial processes, reducing the need for intermediaries and streamlining operations

Security The decentralized and cryptographic nature of blockchain enhances the security of financial transactions, protecting against fraud and unauthorized access

Agility Blockchain facilitates faster and more efficient cross-border transactions, contributing to financial systems' agility

Augmented reality (AR) and virtual reality (VR)

Precision AR/VR can provide immersive data visualization tools, allowing financial professionals to analyze complex datasets with precision

Optimization Virtual banking experiences and AR-driven financial education platforms optimize customer engagement, improving financial literacy and decision-making

Security While not a primary focus in finance, AR/VR applications can enhance security through immersive authentication methods

Quantum mechanics

Precision Quantum computing has the potential to solve complex mathematical problems with unprecedented precision, improving modeling and forecasting

Optimization Quantum algorithms can optimize portfolio management, risk assessment, and complex financial simulations more efficiently than classical counterparts

Security Quantum cryptography can enhance security protocols, providing unbreakable encryption methods to safeguard sensitive financial information

Agility Quantum computing's speed and parallel processing capabilities contribute to financial systems' agility, allowing rapid calculations and decision-making

-making, and language translation. AI's applicability in finance is very broad, ranging from customer service chatbots to complex data analysis and predictive modeling (Luo et al. 2019; Xiao and Ke 2021).

Background of AI

AI) is the process by which machines imitate human behavior. Machine learning is a subset of AI that attempts to analyze and make decisions. Deep learning is a subset of machine learning that uses various neural networks to solve practical problems. AI is the foundation of AI, and machine and deep learning can use AI-related models and mechanisms to solve complex data-oriented problems (Luo et al. 2019).

AI includes various disciplines, such as natural language processing (NLP), computing vision, and expert systems. AI is categorized into three stages: narrow, general,

and super AI. Narrow AI is also known as weak AI. This involves applying AI to specific tasks. Many existing systems that currently use AI have weak AI and focus on narrowly defined problems. They operated within a limited range of predefined functions. Although these may be sophisticated examples of weak AI, they have no real intelligence or self-awareness. Two examples are face identity authentication used in the iPhone and Tesla Autopilot features (Zhou et al. 2022).

Although machines cannot think or reason, the recognition level of general AI is almost the same as that of humans. Machines can perform intelligent tasks using both analytical and computational capabilities. Artificial super intelligence implies that the ability of machines will surpass that of humans. To date, this has only been seen in fiction (Zhang and Kedmey 2018).

Descriptive analysis of AI in finance

We selected 50 articles on AI in Finance. The figure below (Fig. 2) shows the trend of published articles from 2017 to 2023; the overall trend is increasing on an annual basis.

These 50 papers are widely distributed. As shown in Table 3, these disciplines include computing, information systems, engineering, and transportation. It can be seen that AI finance is not purely commercial, but that it is more developed in the directions of science and engineering. FinTech is an interdisciplinary field that employs advanced technologies and skills to solve modern financial problems.

For example, a WordCloud (Fig. 3) was constructed to display the keywords in the 50 FinTech-related papers (by Bibliography Shiny of R). The key terms included AI, machine learning, analytical models, blockchain, data science, economics, mathematics, predictive models, and statistics, etc.

Overview of AI in finance

Recently, the finance sector has undergone a remarkable paradigm shift by integrating AI. This pioneering technology prompted disruptions in traditional financial practices and has yielded implications across domains such as trading, risk assessment, fraud detection, customer service, and personalized financial planning. An exploration of

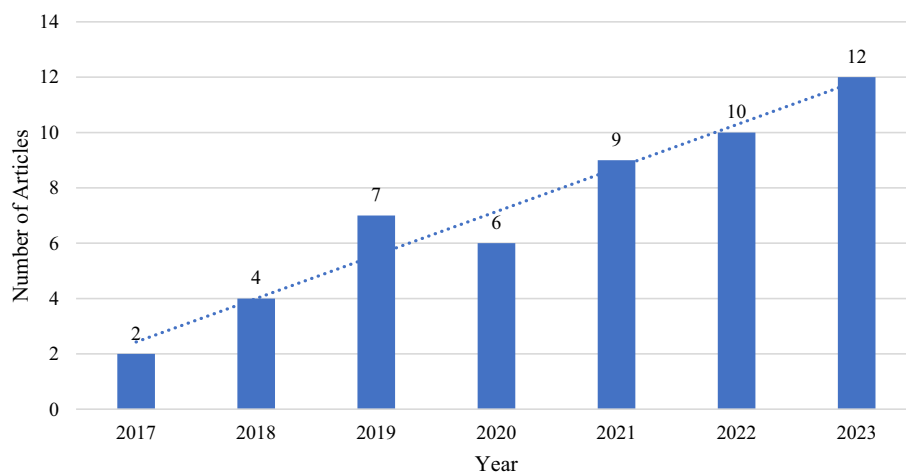
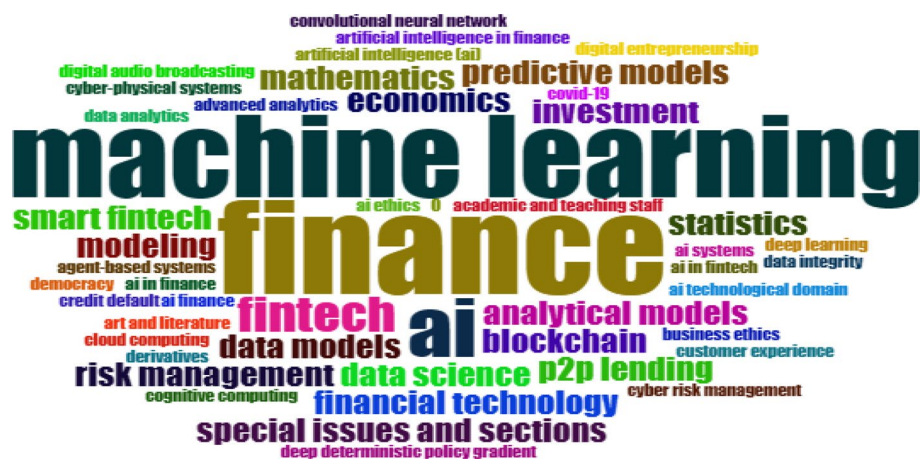


Fig. 2 Year trend of publication trend

Table 3 Publication resources

Sources	Articles
Financial innovation	6
Journal of financial economics	5
The review of financial studies	5
Journal of financial and quantitative analysis	4
California management review	3
MIS quarterly	3
Harvard business review	2
Journal of management information systems	2
Frontiers of information technology & electronic engineering	2
Management science	2
Journal of business research	2
International journal of information management	1
ACM computing surveys	1
Benchmarking-an international journal	1
Computational intelligence and neuroscience	1
E-Mentor	1
Frontiers in artificial intelligence	1
IEEE multimedia	1
International journal of data science and analytics	1
Journal of Asian finance economics and business	1
Journal of investment management	1
Journal of sustainable finance & investment	1
Mathematical problems in engineering	1
Scientific programming	1
Transportation research part E-logistics and transportation review	1

**Fig. 3** High frequency of keywords

the influence of AI in finance underscores its potential, which is nothing short of being inspiring (Fountain et al. 2019; Karolyi and Van Nieuwerburgh 2020; Hu et al. 2023).

One of the most noteworthy applications of AI in the financial sphere is its pivotal role in algorithmic trading. AI-infused trading systems have increased efficiency

and rapidity to unprecedented heights, enabling the real-time execution of intricate strategies while concurrently curbing instances of human error. The employment of machine learning algorithms to scrutinize extensive repositories of historical data, market trends, and news events has led to the identification of lucrative patterns and the optimization of investment decisions. This outcome yielded tangible benefits for financial institutions, hedge funds, and retail traders, manifesting as augmented returns and diminished market volatility (Chen et al. 2019; Fedyk and Hodson 2023).

Moreover, AI is revolutionizing risk assessment. Traditional risk models, which are often plagued by challenges in keeping pace with the ever-evolving financial landscape, have unintended consequences during times of economic flux. In contrast, AI-powered risk models leverage insights driven by data analysis to anticipate potential risks, enhance informed decision-making by financial institutions, and mitigate potential losses (Busse et al. 2020; Engle and Campos-Martins 2023).

Significant advancements have also been made in fraud-detection owing to the influence of AI. Machine learning algorithms are now equipped with the capability of discerning patterns indicative of fraudulent activities with an unprecedented degree of accuracy. This development translates into minimized losses for both financial institutions and their clientele. The proficiency of AI systems in the real-time analysis of vast transactional datasets facilitates the prompt identification of suspicious activities and fortifies defense against unauthorized access (Chen et al. 2019; Lou and Wu 2021).

Notably, AI has substantially transformed customer service in the financial sector. This transformation was spurred by the advent of AI-powered Chatbots and virtual assistants, which provide personalized and real-time support to customers to augment the quality of the user experience and mitigate the need for protracted waiting times. Integrating AI-driven customer service has emerged as a paramount asset for financial institutions in their pursuit of cultivating enduring customer relationships and enhancing brand loyalty (Chen et al. 2019; Karolyi and Van Nieuwerburgh 2020).

Furthermore, the influence of AI extends to the transformation of personalized financial planning and wealth management. Through an analysis of individual spending behaviors, risk-tolerance thresholds, and financial aspirations, AI algorithms are uniquely poised to offer bespoke investment counseling and strategies for wealth management. This degree of tailored personalization, heretofore beyond reach on such a scale, has democratized access to premium financial advice, rendering it accessible to investors from diverse backgrounds (Goldstein et al. 2021; Engle and Campos-Martins 2023; Zhang et al. 2023a, b).

However, addressing the challenges and concerns are intrinsic to the intersection of AI and finance. The reliance on intricate algorithms and extensive datasets leads to queries regarding transparency and accountability. In instances where AI is vested with decision-making authority, the elucidation of the mechanisms underpinning these decisions becomes pivotal, particularly when the financial implications are profound. Additionally, anxieties surrounding data privacy and security are accentuated within an AI-driven financial ecosystems. Safeguarding sensitive financial data from potential cyber threats is crucial (Traczynski 2017; Barroso and Saxena 2022; Alp Coşkun et al. 2023).

Despite these challenges, the influence of AI on financing is irrefutable. Its faculty for task automation, amplification of decision-making acumen, and capacity for innovative

problem-solving have indubitably transformed the financial landscape. As this technology continues its maturation, complemented by regulatory endeavors to mitigate potential risks, the horizon for AI in finance appears promising (Lou and Wu 2021; Grennan and Michaely 2021).

The integration of AI within the financial sector heralded a new era defined by innovation, operational efficiency, and client-centric services. Its contributions, spanning algorithmic trading, risk assessment, fraud detection, customer service, and personalized financial planning, are reconfiguring interactions with financial institutions. Navigating this evolving landscape attests to AI's characterization as not merely a tool, but also a potent driving force steering the metamorphosis of finance, as conventionally perceived. The responsible and ethical embracement of AI remains pivotal for fully unlocking its potential and reaping its manifold advantages for both the industry and its stakeholders (Busse et al. 2020; Trieu et al. 2022).

Applications of AI in finance

As a major application, AI-related technologies penetrate broad aspects of finance. Examples (Table 4) include NLP, trust, education, pandemic-related issues, ethics, estimation, and review.

Review articles

The development cycle of AI and related technologies is still very long, and there are still many phenomena that cannot be explained or studied (Zhou et al. 2022), that is, they are the black box of machine learning models. One study focuses on startups, which are AI-related financial companies in various fields such as finance, data analysis, robotics, insurance, and marketing (Zhang and Kedmey 2018). Another study proposes an AI-based financial framework that relies on technology at various times. Many problems

Table 4 Applications of AI in finance

Application	Explanations
NLP (national language process)	As a major technique of artificial intelligence, NLP (National Language Processing) is used to build a system to find financial terms from languages, e.g., Vietnamese. This type of tool can provide valuable financial information and can track underlying trends in financial markets (Pham, et al. 2022)
Trust	Comparing AI and human trust performance by simulating and experimenting with financial activity shows that neither trust mechanism alone is sufficient. Rather, human intervention and AI guidance are better decision-making solutions (Maier, Menold, & McComb, 2022)
Education	According to the development of artificial intelligence-related technologies, it is necessary to offer financial courses on the direction of artificial intelligence for college students. An empirical study tests this argument and claims that the curricula of finance courses can be better modified in response to technological and social developments (Grabirńska et al. 2021)
Pandemic finance	Due to the COVID-19 pandemic, the entire financial world has been affected (Ho 2023). Given a pandemic, not just the COVID-19 virus, we have had to adapt to changing circumstances. AI and other technologies may lead to stronger, more resilient solutions
Ethics	AI is a high-performance advanced technology; on the other hand, individual response to AI seems to be an ethical issue. A well-designed AI-oriented system infrastructure can promote financial sustainability (Brusseau 2021)
Estimation	While it is still difficult to demonstrate the performance of these methods, this study is a good attempt to use AI-related assessments in operations, entrepreneurship, customer relationship management, decision support systems, and more (Luo et al. 2019; Arora, et al. 2023)

can be solved using advanced technologies such as AI (Zheng et al. 2019). Two studies provide in-depth overviews of AI in finance, including economic and financial challenges, related data issues, and research on AI in finance (Cao et al. 2021; Cao 2022).

Machine learning for finance

Machine learning is a branch of AI that can help manage assets and evaluate risks by making accurate predictions. Machine learning systems are algorithms that can learn from data, identify patterns, and make decisions. Analyzing vast amounts of data and developing statistical models are essential for optimizing financial strategies. Leveraging the power of AI can improve performance in several financial applications. Potential improvements range from implementing cost-saving solutions to developing a unique competitive advantage. Its adaptability and ability to uncover patterns in large datasets make it a crucial area of study (Kolchinsky 2018; Huang and Shi 2022; Yae and Luo 2023).

Background of machine learning

Life experiences represent the gradient of human development. We build machines (robots) and we program them to the best of our knowledge. The ideal is for machines to learn, process, and decide what they can get out of the world. This is known as machine learning. The first step involves learning, and, more importantly, understanding and reasoning. Machine learning learns the data and builds prediction models. The more data analyzed, the better the performance of the model, and the higher the accuracy of the prediction. Machines learn in several ways, and can be supervised, unsupervised, or reinforcement (De Spiegeleer et al. 2018; Lin and Bai 2022). The following table (Table 5) presents basic interpretations and examples.

Supervised learning

Supervised learning has clear goals because of the labeling and labeled data. Through the training process, it was possible to obtain the expected results. Computers learn to model relationships based on labeled training data. The inputs and outputs are labeled for the algorithm. After learning how best to model the relationships for the labeled data, the training algorithms were used to model or predict the outcomes for the new datasets. For example, algorithms can be used for regression analysis to predict stock market returns and revenue growth; for classification, generating buy, sell, and hold signals; the likelihood of a successful M&A or IPO; to enhance the detection of fraud in financial statements; or for the classification of winning and losing funds or ETFs (Osi-sanwo et al. 2017; Singh et al. 2016).

Table 5 Classification of machine learning

Categorization	Features	Model
Supervised learning	Labeled data and learning	K-NN, Regression, Classification, Naive Bayesian Model, Random Forest Model, Support Vector Machines
Unsupervised learning	Unlabeled data and learning	Ordination, clustering, anomaly detection, neural networks, K-Means, PCA
Reinforcement learning	Rewards incentives and learning	Markov Decision Process, Q Learning, DQNS

Unsupervised learning

In this case, the raw data do not contain labels. The goal is to discover the underlying patterns and identify groups of samples that behaved similarly. For example, for clustering, data may be grouped on non-standard characteristics such as financial statement data or corporate characteristics rather than geographical territory, client profiling and asset allocation, and portfolio diversification and stock selection, based on co-movements similarities; for dimensionality reduction, data may identify the most predictive factors underlying asset price movements (Mahesh 2020; Alloghani et al. 2020).

Reinforcement learning

Reinforcement learning is a reward-motivated learning process involving a feedback loop. The training process for the model input data is regular. If the results are satisfactory, the process ends. Otherwise, the process iterates over the training data based on the feedback until a satisfactory result is obtained. In reinforcement learning, a computer (agent) learns by interacting with the environment, producing actions, and discovering rewards. Environment, actions, and reward systems must be defined. The machine then explores and exploits to maximize the reward. New actions may not be immediately optimal. Subsequently, learning occurs through millions of trials and errors. For example, a virtual trader (agent) follows certain trading rules (actions) in a specific market (environment) to maximize its profits (reward) (Arulkumaran et al. 2017; François-Lavet et al. 2018).

Descriptive analysis of machine learning for finance

Among the selected 96 papers, approximately 85.41% were published in the last four years: 2020 (11), 2021 (17), 2022 (25), and 2023 (29), as shown in Fig. 4. With the development of finance and data analytics, machine learning is becoming increasingly important and is attracting increasing attention.

Table 6 lists the top 20 sources that publish machine learning-related financial papers. This distribution is similar to that of AI-related finance articles, which comprise articles from various disciplines.

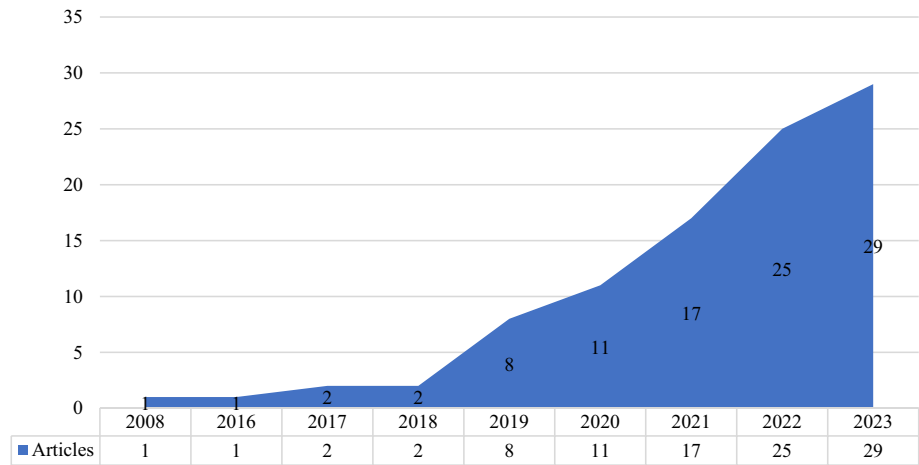


Fig. 4 Annual publication trend

Table 6 Top 20 publication resources

Sources	Articles
Financial innovation	15
Journal of financial economics	8
MIS quarterly	8
The review of financial studies	6
Journal of financial and quantitative analysis	5
The journal of finance	4
Annals of operations research	4
Computational intelligence and neuroscience	4
Neural computing & applications	3
Quantitative finance	3
Applied sciences-basel	2
Entropy	2
Risks	2
Management science	2
Finance research letters	2
Journal of behavioral finance	1
Manufacturing & service operations management	1
International journal of production economics	1
Abacus-A journal of accounting finance and business studies	1
Annals of financial economics	1

**Fig. 5** High frequency of keywords

From a WordCloud of keywords (based on Bibliography Shiny of the R Project), we can see that machine learning has been applied to many financial activities such as supply chain finance, credit risk forecasting, quantum machine learning, venture capital, anti-money laundering, and asset pricing (Fig. 5).

Overview of machine learning in finance

Machine learning has surfaced as a transformative technological paradigm, instigating revolutionary changes across diverse industries, and the finance sector is no exception. We embarked on an exploration of extraordinary advances catalyzed by machine learning within the financial domain. We examine its applications, advantages, challenges,

and future directions (Bernard et al. 2020; Chau et al. 2020; Griffin et al. 2021; Depren et al. 2021; Cheng et al. 2023).

The infusion of machine learning into the finance fabric has several advantages: (1) Augmented Efficiency. The automation of repetitive tasks coupled with expedited data analysis furnishes financial institutions with heightened operational efficiency while concurrently curtailing the prevalence of manual errors. (2) Enhanced Decision-making. Machine-learning algorithms, adept at processing voluminous datasets, provide insights and predictions that empower the informed and timely formulation of decisions. (3) Mitigating risk through accurate risk prediction and the identification of aberrations, machine learning diminishes financial perils, thus contributing to the establishment of more robust and resilient financial ecosystems. (4) Economies of Scale. The confluence of automation and streamlined processes translates into tangible cost efficiencies, rendering financial services more accessible and cost-effective for an expansive clientele (Shin et al. 2020; Huang et al. 2021; Erel et al. 2021).

Notwithstanding its remarkable potential, machine learning within the financial ambit confronts a series of pertinent challenges: (1) Data Privacy and Security. The handling of sensitive financial data requires the implementation of stringent security measures to prevent potential breaches and cyber intrusions. (2) Model Interpretability. Many machine learning models exhibit a complex architecture, that potentially hinder the comprehension of their decisions, which is an imperative aspect within a heavily regulated domain such as finance. (3) Algorithmic Overreliance. An undue dependence on machine-learning models might inadvertently sideline the indispensable role of human judgment, especially in contexts that necessitate nuanced and contextual decisions. (4) Bias and Fairness. Machine learning algorithms trained on historical datasets may inadvertently perpetuate latent biases, thus resulting in the differential treatment of specific demographic groups or the reinforcement of existing disparities (Grushka-Cockayne et al. 2017; Li et al. 2021a; Kaniel et al. 2023).

The trajectory of machine learning within financial expanses portends a future characterized by immense promise: (1) Explainable AI. Advancements in explainable AI are poised to address the challenge of model interpretability and to confer upon financial institutions an enhanced understanding of the rationale underlying machine learning-based decisions. (2) AI-Driven Regulatory Compliance. Machine learning is poised to support financial institutions in their adherence to evolving regulatory mandates, accomplished through the automation of compliance processes and mitigation of human-induced errors. (3) Reinforcement Learning in Trading. The progressive evolution of reinforcement learning algorithms is primed to catalyze the augmentation of algorithmic trading strategies, thereby optimizing portfolio management and ameliorating risk exposure (Sturm et al. 2021; Meursault et al. 2021; Özlem and Tan 2022; Griffin et al. 2023).

Machine learning has galvanized a transformative tide within the financial landscape, serving as an impetus for innovation, operational efficiency, and risk management. Although challenges persist, an unwavering commitment to ongoing research and development holds the promise of fully unlocking the latent potential of machine learning and constructing a finance domain characterized by data-driven paradigms, fortified security, and an overarching focus on customer -centricity. As this odyssey

reveals, the convergence of AI and finance augurs a future marked by a more luminous and inclusive financial landscape, benefiting all stakeholders involved (Li et al. 2021b; Kane et al. 2021; Sautner et al. 2023).

Applications of machine learning in finance

Machine-learning algorithms are employed by many firms for fraud prevention, risk management, wealth and investment management, customer service underwriting, and trading. Fintech companies can quickly utilize and implement machine learning during their operations. Traders use machine learning to improve their market predictions by developing algorithms and setting instructions for certain parameters (Gunduz 2021; Zhang et al. 2023a, b). Machine learning is an important tool for promoting the accuracy and performance of financial systems (Chen and Tsai 2020). Its applications (Table 7) described in the research are diversified and include entrepreneurial finance, health, challenges, Internet finance, money laundering and terrorist financing, Internet finance, agriculture, risk control, supply chain finance, quantitative finance, and reviews.

Table 7 Applications of machine learning in finance

Application	Explanations
Entrepreneurial finance	Machine learning technology could be an innovative way for startups or investors to chase and to compete in the market (Ferrati and Muffatto 2021; Ozgur et al. 2021)
Healthy	Along with another technique, NLP (National Language Processing), appropriate machine learning algorithms are employed to track and estimate health finances (Dixit et al. 2022)
Challenges	It is impossible to use machine learning for finance to offer all benefits without any drawbacks. How we balance and mitigate potential consequences is becoming an important factor influencing the development of financial machine learning (Kolchinsky 2018; Hansen and Borch 2021; Antunes 2021)
Money laundering and terrorist financing	Machine learning algorithms are one way to investigate illegal activities in the banking industry, such as money laundering and terrorist financing. However, due to limited information and an unproven basis, there is still a long way to go to find well-designed and powerful machine learning models to track and protect banking and financial systems (Canhoto 2021; Prisznyák 2022; Gupta and Pierdzioch 2023)
Internet finance	Another trend in finance is development through the Internet; with it comes uncertainty and unexpected phenomena. The internet can provide rich data, and machine learning models are durable to analyze the financial Internet, credit risk, and financial audits, using tools such as the Apriori algorithm and the Bayesian network (Ma and Lv 2019)
Agriculture	Machine learning is related with agriculture, such as agricultural logistics financing (Yang 2019) and supply chain finance for agricultural SMEs (Belhadi et al. 2021)
Risk control	A well-designed machine learning algorithms detects corporate financing risks to find solutions. Datasets combined with machine learning algorithms may not monitor financial risk as precisely as intended. A good algorithm should indeed be practical for dealing with financial risk problems (Zhu et al. 2017; Alonso and Carbo 2022; Ma et al. 2023)
Supply chain finance	In the discipline of supply chain and logistics, machine learning models are used to monitor and to optimize the potential value of finance, such as the overall framework, payment infrastructure, risk assessment of SMEs, etc. In addition, blockchain, IoT (Internet of Things), and other technologies complement each other to improve the performance of supply chain financial machine learning algorithms (Zhu et al. 2016, 2019)

Quantitative finance

Of the selected papers, 12 focused on machine learning-based quantitative finance. We present a table (Table 8) to explain the main content of the paper, such as the algorithm used and the research purpose.

Review articles

Support Vector Machines (SVM), Artificial Neural Networks (ANN), and Genetic Algorithms (GA) are popular machine learning algorithms used in energy economics and in finance (Ghoddusi et al. 2019). SVM, Random Forest (RF), and DL models (LSTM or RNN) outperform traditional models for malicious data and quantitative financial estimates (Rundo et al. 2019; Ni et al. 2021).

Israel et al. (2020) illustrate why and how machine learning mechanisms are adopted in finance. The reviews included financial topics related to machine learning algorithms. Three topics are attractive: (1) investment analysis, (2) asset modeling and forecasting, and (3) risk management (Aziz et al. 2022). One study explored detection to analyze emerging themes in FinTech (Cai et al. 2019). Another study describes sustainable financing and emerging technologies (Kumar et al. 2022). Asset pricing, stock market returns, bankruptcy prediction and risk credit scoring, anomaly detection in financial reporting, and popular algorithms are also target topics in related research (Almaskati 2022; Htun et al. 2023).

Additionally, several studies have performed bibliometric analyses to explore the link between machine learning and finance. Certain tracks reflect value, such as (1) Portfolio Construction, Valuation, and Investor Behavior; (2) Financial Fraud and Financial Distress; and (3) Sentiment Inference, Forecasting, and Planning (Goodell et al. 2021; Warin and Stojkov 2021; Ahmed et al. 2022).

Table 8 Machine learning algorithms in quantitative finance

Algorithms	Objectives	Paper
Gaussian process regression, machine learning algorithms	Speedup, Accuracy	De Spiegeleer et al. 2018
The supervised adaptive group LASSO (SAGLasso) method	Return Estimation, Transparency, Accuracy	Huang and Shi 2022
Restricted Boltzmann machines (RBMs), quantum circuit Born machines (QCBMs)	Comparison, between conventional and quantum machine learning	Alcazar et al. 2020
The DEA method	Analyzing factors impacting performance	Liu and Zhan 2019
Convolution neural network (CNN)	Analyze, distribute, and duplicate finance data	Zhao 2021
The XGBoost method	Featuring and analyzing debt credit	Lin and Bai 2022
Machine learning sentiment analysis	Investigating participants' financial behavior	Hansen and Borch 2022
Deep learning algorithm	Asset Pricing of Average Option	Gan et al. 2020
Stacked generalization and boosted trees	Analyzing factors of venture capital	Taboga 2022
Principal component analysis (PCA), K-nearest neighbors (KNN), and Artificial neural networks (ANN)	Performance of working capital finance	Mahmood et al. 2022
Text-mining and random forests	Identifying social factors impacting financial return	Gonzales Martínez 2020

Blockchain in the financial revolution

Blockchain technology integrates data processing, security encryption, computing algorithms, and consensus mechanisms. Blockchain enables transactions between participants to become cost-effective and convenient. The concept of cryptocurrencies, or virtual currencies that run on a technology called blockchain, has emerged. Hackers will not be able to change the data in the blockchain for long periods. Every user has a copy of the ledger, and the data in the blocks are encrypted using complex algorithms. A blockchain can be described as a collection of interrelated and highly tamper-resistant records. Blockchain is a decentralized and distributed ledger technology that ensures secure, transparent, and tamper-resistant record -keeping. In finance, blockchain is synonymous with cryptocurrencies such as Bitcoin and has the potential to revolutionize transactions, settlements, and smart contracts. Its focus on transparency and security makes it an essential technology for investigations (Sebastião and Godinho 2021; Ma et al. 2023).

Background of blockchain

Blockchain technology is an innovative way to achieve decentralization. It is a system that maintains records for each participant without central authorization or intervention. This is a decentralized mechanism for maintaining ledgers that are difficult to falsify. Because participants were involved in justifying the records, it was difficult to break the rules. The blockchain function of the system has four elements. The first element is a peer-to-peer network, which is a network of computers with equal privileges, also known as a node. A network is necessary to allow each participant to access the system, communicate, and share information. The second aspect is cryptography. Cryptography builds a relatively secure environment to protect underlying transactions. Most had to verify the authentication of each participant. The third element is a consensus algorithm. In a blockchain, many consensus mechanisms, such as Proof of Work (PoW), Proof of Stake (PoS), Proof of Activity (PoA), and Proof of Importance (PoI) facilitate messages to the chain. Finally, punishments and rewards are provided. This ensured that all participants followed the rules equally at all times. This reward is a token or coin awarded every time a consensus is reached and a new block is added to the chain (Choi 2020a).

Blockchains are divided into three categories according to the different levels of permissions and accessibility requirements: public chains, consortium chains, and private chains. The public chain allows each participant access without a threshold, the consortium chain allows a relatively small number of participants access through appropriate authorization, and the private chain is limited to a few participants with special requirements. Blockchains are programmed to have many features that could benefit financing such as decentralization/distribution, encryption, immutability, security/privacy, consensus, and transparency (Harwick and Caton 2022).

Descriptive analysis of blockchain in finance

As a popular emerging technology, blockchain has been applied in the financial field for over a decade, and an increasing number of studies explored the internal connections between blockchain and financial systems. The graph below (Fig. 6) clearly shows

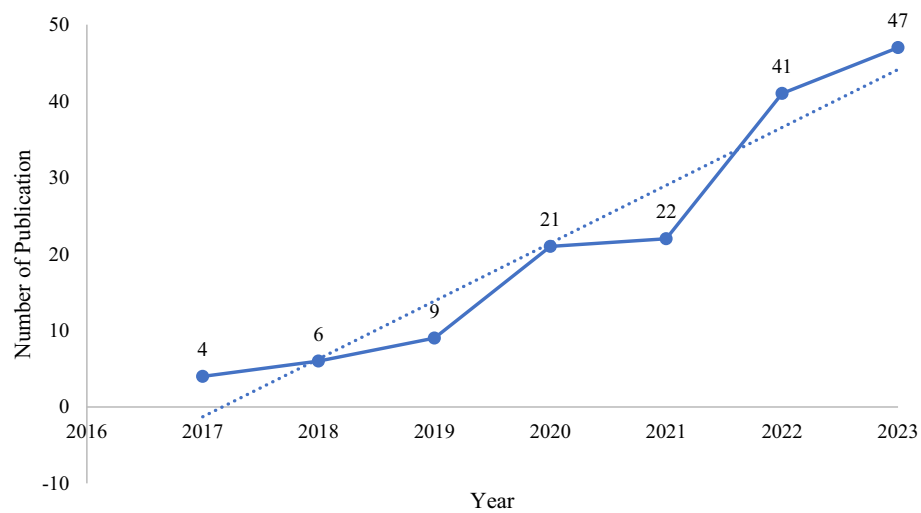


Fig. 6 Year trend of publication

Table 9 Top 20 resources

Sources	Articles
Financial innovation	18
The review of financial studies	9
Journal of financial economics	7
IEEE transactions on engineering management	6
International journal of production research	5
Accounting & finance	4
MIS quarterly	4
Annals of operations research	4
Frontiers of engineering management	3
International journal of production economics	3
Journal of banking and finance law and practice	3
Journal of enterprise information management	3
Journal of theoretical and applied electronic commerce research	3
Strategic change-briefings in entrepreneurial finance	2
Technological forecasting and social change	2
Technology analysis & strategic management	2
The journal of finance	2
Journal of financial and quantitative analysis	2
International journal of information management	1
Annals of financial economics	1

that many papers are blockchain-related financial research; therefore, we selected 150 papers for review. Of these, 41 and 47 of the 150 papers were published in 2022 and 2023, respectively.

Because blockchain is a major application technology in the financial field, related research in high-quality journals provides better insights and arguments (Table 9). These are Financial Innovation, IEEE Transactions on Engineering Management, the International Journal of Production Research, the International Journal of Production

Economics, Financial Innovation, the Journal of Enterprise Information Management, Technology Forecast and Social Change, and so on.

A WordCloud of the keywords (Fig. 7, based on Bibliography Shiny of the R Project) highlights a few relevant topics, such as supply chain finance, smart contracts, crowd-funding, and cryptocurrency. In the next section, most applications are described in detail.

Overview of blockchain in finance

In recent years, the blockchain technology has fostered a transformative force. These decentralized, immutable, and transparent features have the potential to fundamentally reshape the contours of the financial landscape, thus meriting recognition as veritable game changers. We explore the repercussions of the incorporation of blockchain within finance, tracing its trajectory as it paves the way for an inherently more efficient, secure, and all-encompassing financial ecosystem (Griffin and Shams 2020; Cong et al. 2022).

Foremost among the salient benefits of blockchain technology in finance is the promotion of enhanced security. Traditional financial systems have long grappled with persistent issues of fraud, data breaches, and cyber intrusions, culminating in substantial losses for both financial institutions and clients. The unique attributes of blockchain, wherein each transaction is meticulously recorded on an immutable ledger distributed across a network, engender a state of near -impenetrability against manipulation or alteration of data by malicious entities. This inviolable nature imbues a sense of trust and assurance, thus magnetizing a larger user base to engage in financial undertakings (Hinzen et al. 2022).

The transparency introduced by blockchain is equally significant. Historically, financial processes have been frequently enshrouded in opacity, leaving customers uncertain about the management of their financial resources. With the advent of blockchain technology, every participant in a network gains unfettered access to the entire chronicle of transactions, thereby facilitating a climate of openness and accountability. This new-found transparency emerges as a consequence of ameliorating regulatory compliance, as auditors are empowered to seamlessly trace financial activities and secure alignment with pertinent statutes and regulations (Cai 2018; Amoussou-Guenou et al. 2023).

Moreover, the elimination of intermediaries from numerous financial processes is a watershed moment. Traditional actors such as banks and payment processors have long



Fig. 7 High frequency of keywords

introduced complexities and associated costs into transactions. Blockchain architecture facilitates peer-to-peer transactions and avoids the necessity for intermediaries and the attendant fees they impose. This disintermediation reduces costs and expedites transactional velocity. Blockchain enables real-time settlement and curtails the latency endemic to cross-border payments through more advanced mechanisms (Saleh 2021; Saleh 2021).

However, the influence of the blockchain transcends the limitations of conventional financial services. It has precipitated decentralized finance (DeFi), an expansive and swiftly burgeoning ecosystem offering a spectrum of financial services—stripped of the traditional intermediaries that hitherto characterized these processes. Constructed using blockchain protocols, DeFi platforms have democratized financial prospects for millions of unbanked and underbanked individuals worldwide, fostering financial inclusivity on an unprecedented scale (Cong and He 2019; Miranda et al. 2022).

Nonetheless, it is imperative to extend cognizance toward the realization that the full potential of blockchain within the financial sphere is not a challenge. The overarching concern pertains to scalability as blockchain networks confront the task of accommodating an escalating number of transactions while upholding efficiency. Additionally, the conundrum of harmonizing data privacy with the maintenance of transparency poses a delicate confluence for financial institutions to embrace blockchain-driven solutions (Ilk et al. 2021).

The assimilation of blockchain technology within the financial industry has instigated a seismic reconfiguration of the manner in which financial transactions are conducted. Their heightened security, transparency, and disintermediation capacities have become instrumental in shaping the financial landscape and democratizing access to financial services. Although impediments remain, the potential of blockchain to catalyze a transformative effect on finance remains indisputable. As a growing multitude of industries embrace this innovative technology, we bear witness to the nascent stages of an epoch in finance characterized by enhanced inclusivity, heightened security, and unparalleled operational efficiency (Chiu and Koepl 2019; Chod and Lyandres 2023).

Applications of blockchain in finance

Since the launch of Bitcoin, blockchain has been applied in many fields (Table 10), such as Agroindustry, Climate, Cryptocurrency, Electricity, Entrepreneurial Finance, Healthcare, ICO, Data Processing, Infrastructure, Insurance, Invoice, Tourism, Trade finance, Risk Control, Decentralization, Supply Chain Finance, Country Case, and Review Articles. This section will explain the blockchain applications in each relevant field.

Based on the selected papers and relevant disciplines, we summarize the features of the applied blockchain (Table 11). For instance, decentralization and transparency.

Decentralization

Blockchain and smart contracts have the potential to make the contemporary financial systems more effective and efficient (Meralli 2020). Schär (2021) discussed various tokenized cryptocurrencies, consensus mechanisms, and smart contracts for financial markets. A decentralized blockchain financial system is a safe and transparent trading platform. When a transaction or deal is completed, the system provides

Table 10 Applications of AI in finance

Application	Explanations
Agroindustry	A framework for agricultural supply chain finance that has the potential to address practical issues of traditional supply chain systems (Wang and You 2022), such as data management and analysis, logistics performance, and security (Rijanto 2021a)
Climate	Blockchain (distributed ledger technology) has the potential to design an innovative green climate fund in which governments or institutions can facilitate monitoring and managing the environment (Schulz and Feist 2021)
Cryptocurrency	One of popular cryptocurrencies, Ethereum was analyzed and priced by linear regression and SVM (Poongodi et al. 2020; Sebastião and Godinho 2021; Bazzanella and Gangemi 2023)
Electricity	Blockchain is potential to construct special purpose vehicles (SPVs) to solve power needs of electricity that entities have restrained access (Alao and Cuffe 2021)
Entrepreneurial finance	Analyses of social media data over three time periods were employed, looking for links between the elements of the entrepreneur, such as blockchain, cryptocurrencies, and ICO (Initial Coin Offerings) (Bogusz et al. 2020; Kherbouche et al. 2022). Another study empirically examined other aspects, such as real estate investment trusts (REITs) and gender differences (Fisch et al. 2022)
Healthcare	Decentralized, secure, and transparent blockchain and cryptocurrencies can advance global healthcare, especially in the areas of universal health equity and financing (Till et al. 2017)
ICO	For startups, initial coin offerings (ICOs) are critical to their survival and growth (Ali et al. 2023). Cryptocurrency-oriented startups reflect similar patterns in crowdfunding and VC funding (Ante et al. 2018). Compared to traditional banking (TBK) projects, due to the unique characteristics of blockchain and related technologies, start-up ICOs behave differently in terms of pricing, estimation, decision-making, and strategy (Choi 2020a)
Data processing	In the financial system, many activities are related to data analytics and processing, such as information processing (Su et al. 2022), information transparency (Guo et al. 2022; Chod et al. 2022), information sharing (Zheng et al. 2022), and secured and trusted data (Liu et al. 2021a, b). Machine learning is a powerful tool that can assist finance deal with information and data
Infrastructure	Machine learning algorithms execute on well-designed infrastructures (Du et al. 2020). One type of decentralized finance is token-oriented financial infrastructure, which is still in the early stages of development (Tian et al. 2020). Another study compares financing structures with blockchain-based business performance in the context of IoT financial models (Li, et al. 2022)
Insurance	A blockchain-based insurance was designed to implement orphan drugs and to reduce high prices (Ho et al. 2022)
Invoice	A public blockchain-based platform of invoicing plays the important roles of sharing records, preventing sensitive information, and securing transactions (Guerar et al. 2020; Priem 2020; García-Corral et al. 2022)
Tourism	For the tourism industry, among cryptocurrency-oriented ICOs, Bitcoin is still the most incentivized instrument to enrich the foundations (Faghih Mohammadi and Heidari 2020; Bulut 2022)
Trade finance	The performance of trade finance could be lifted by blockchain in the aspects of data exchange and communication, trust between participants, and information security (Kowalski et al. 2021; Rjoub et al. 2023; Bouri et al. 2023)
Risk Control	Blockchain has the potential to establish a secure international e-commerce transaction system. Other studies consider risk control in supply chain finance; for example, risk assessment and management models (Dang et al. 2022), accessibility and storage models (Li et al. 2023), and machine learning and the blockchain mechanism (Feng et al. 2023)

token rewards to an entity (Kumar and Amin 2022). Another study discusses autonomous decentralized financial systems from the perspectives of trust and market institutions. One argument might be that the decentralized financial system can be autonomous, and that off-chain financial entities and activities can complement the decentralized financial system (Harwick and Caton 2022).

Table 11 Blockchain features in different disciplines

Discipline	Features of blockchain
Agroindustry	Traceability, storage, security, transparency,
Climate	Transparency, DLT, security
Cryptocurrency	Decentralization, DLT
Electricity	Decentralization, security, intelligent
Entrepreneurial finance	Initial coin offerings (ICOs), consistency, integrity, and security
Healthcare	Storage, security, information disclosure
ICO	Decentralization, cryptocurrency
Infrastructure	Tokenization, security, smart contract, Hyperledger Fabric, interoperability
Insurance	Authentication, security, cost-saving
Invoice	Encryption, anonymous, security, transparency, immutable, smart contract
Risk control	Convenient, security, encryption
Supply chain finance	Decentralization, security, transparency, smart contract, immutable, encryption
Tourism	ICO, cryptocurrency

Supply chain finance

Blockchain is a core technology used to create direct and invisible value in supply chain finance (Wang and Xu 2023; Ning and Yuan 2021; Gong et al. 2022a, b). The Technology Acceptance Model (TAM) is used to judge the ease of use and perceived usefulness of blockchain in supply chain finance. The results are optimistic, with limited data (Rijanto 2021b).

As a disruptive technology, blockchain has the potential to serve supply chain systems at different levels. This is a good attempt to track blockchain in supply chain finance; however, many questions remain (Dong et al. 2022a, b; Kucukaltan et al. 2022; Liu et al. 2021a, b). In emergency situations, such as during the COVID-19 pandemic, supply chain financial systems are also affected. Advanced-designed blockchain supply chain finance might perform differently (Paul et al. 2022).

A comparison, from the perspective of participants of the system, was explored between traditional supply chain finance and blockchain-based supply chain finance. Blockchain technology could benefit supply chain finance in the aspects of securitization, transparency, and optimization (Hofmann et al. 2017; Dong et al. 2021; Yang et al. 2022).

Additionally, supply chain finance research is related to industries, such as automobiles, e-commerce (Harish et al. 2021; Deng et al. 2023), fashion products (Choi 2020b), food (Natanelov et al. 2022; Hong 2021), logistics (Fu et al. 2022), retail (Li et al. 2020a, b), and SMEs (Li et al. 2020a, b; Su et al. 2022; Jiang et al. 2022), etc. A common issue across industries is the adoption of machine-learning models to improve the overall financial performance of related supply chain systems.

Country case

Blockchain-based financing performs differently across geographical territories. Studies include blockchain applications in Islamic finance (Chong 2021), the security structure of finance in Africa (Campbell-Verduyn and Giumelli 2022), the supply chain finance infrastructure for SMEs (Chen et al. 2021a, b), and the Beibu Gulf Region Supply Chain Finance in China (Wang and Wu 2021), distributed ledger technology governance for

banking and finance systems (Simpson 2018) and debt capital markets in Australia (Lunardi 2019), blockchain-based trade finance governance (Ibrahim and Truby 2021) and trade finance in Qatar (Dahdal et al. 2020), blockchain-based supply chain finance in India (Kaur et al. 2022), blockchain supply chain finance in Malaysia (Nor et al. 2021), and blockchain supply chain finance to Ghanaian SMEs (Asante Boakye et al. 2023), among others.

Review articles

Blockchain addresses banking-related issues, such as cost reduction, advantages, risks, future trends (Osmani et al. 2021), supply chain finance (Gong et al. 2022a, b), opportunities and policy implications (O'Dair and Owen 2019), and challenges and responses (Zhang et al. 2020; Fang et al. 2022). Studies concentrate on the challenges and limitations of applying blockchain technology to banking and financial systems (Gan et al. 2021; Jaiswal et al. 2022).

Challenges of enabling emerging technologies in finance

Even with the emerging technology itself, there are many problems to be solved. Enabling emerging technologies in the financial system presents additional challenges owing to the development stage of the technology and the interoperability of emerging technologies across different systems. This study addresses technical issues in financial systems (appropriateness, interoperability, explainability, security, and sustainability), technological challenges, and operational standpoints (Zhou et al. 2022; Kumar et al. 2022). Addressing these challenges requires collaborative efforts among industry stakeholders, ongoing R&D, effective regulatory frameworks, and a strategic approach to technological adoption that balances innovation with risk management.

Technical issues in finance

Do emerging technological solutions fit financial systems' structures and operations? However, there is no clear answer to this question. For example, many studies use machine learning-based models to estimate financial markets to optimize potential profits; however, no one model or framework is superior. These models are still based on traditional financial infrastructure. Emerging technologies can create intangible value through financial activities, and common models and foundations can be employed to enhance the functioning of emerging technologies in finance (Ghoddusi et al. 2019; Cao 2022).

Is the market incentivized to implement emerging technologies and embed them in the existing market infrastructure? In theory and technology, emerging technologies are good for financing; however, it is better to think about this differently. For example, blockchain is a technology suitable for supply chain financing; however, its operating and transaction costs are high, and potential investors and participants are unwilling to promote blockchain or other technologies to cope with practical problems (Paul et al. 2022).

Emerging technologies can create value. However, the interpretation of the intrinsic and extrinsic relationships remains controversial. For example, machine learning or deep learning algorithms can analyze big data and obtain better results. However, black boxes

still exist, and are difficult to track and interpret. Managers do not know enough about the tools, terminology, and potential pitfalls of machine learning to be effective. Any result is commercially useful only if investors, employees, and customers can interpret it (O'Dair and Owen 2019; Sharma et al. 2020).

Security and Privacy Issues. Cybersecurity is an important aspect to consider, particularly in financial systems. For example, artificial- and blockchain-oriented mechanisms can solve related problems and improve the overall security of the financial system. Advanced algorithms are able to detect attacks and false transactions. Blockchain implements cryptographic protocols to secure both the participants' information and the financial system. These benefits can be improved using more advanced and intelligent financial platforms (Zhang et al. 2020; Almaskati 2022).

Sustainability of the financial system. Emerging technologies predict a bright future for financial systems. On the other hand, the sustainable development of the financial system is essential. For example, AI or blockchain could lead to gender or racial bias from the perspective of engineers, employees, and participants, or to related variables that are highly correlated with gender or race in machine learning. Blockchain can help promote inclusive finance, though avoiding the deviation brought about by blockchain is key (Goodell et al. 2021; Gan et al. 2021).

Technological challenges in finance

Integrating diverse emerging technologies often requires compatibility standards to ensure seamless communication and data exchange among the systems. As financial institutions adopt these technologies, ensuring scalability to handle growing data volumes and user interactions has become a critical concern. Dependence on accurate and high-quality data is important for technologies like AI and machine learning, and challenges in data sourcing and cleaning can hinder their performance. Blockchain and quantum technologies face security challenges, including quantum-resistant cryptography and facilitating the integrity of distributed ledgers (Chaklader et al. 2023).

Operational issues in finance

Navigating evolving regulatory issues, particularly in the context of emerging technologies, requires continuous adaptation to compliance standards. The shortage of professionals proficient in emerging technologies poses challenges in building and maintaining competent teams for development and implementation. Overcoming resistance and fostering user trust in new technologies, especially those involving sensitive financial data, requires effective change management and user education. The initial investment and operational costs associated with implementing emerging technologies may be high, and organizations must carefully assess the return on investment. The ethical use of technologies, privacy concerns, and adherence to legal frameworks are critical aspects that financial institutions must navigate for responsible deployment (Chaklader et al. 2023).

Discussions and future directions

In addition to these three major technologies in finance, other technologies impact the financial system, allowing it to be more immersive, accurate, and advanced. These practical scenarios include immersive banking systems, financial inclusion, and quantum financial systems.

Immersive banking system

AR projects computer-generated enhancements onto reality to help perform tasks more efficiently. Halfway between real and virtual reality, AR renders and visualizes real-world data. Therefore, virtual elements resemble the current reality to a certain extent. AR has been in vogue in recent years and the revolution has never stopped. AR empowers customers to provide unique experiences and benefits. AR overlays digital information in the real world, whereas VR immerses users in a simulated environment. Although not as prevalent in traditional financing, AR and VR have applications in virtual banking and immersive financial education. The study of these technologies is motivated by their potential to reshape the customer experience and financial interactions (Marakarkandy et al. 2017).

What is immersive banking? Augmented banking includes open, cognitive, and automated banking. Immersive banking involves advancing the exponential growth of technology to provide a richer experience through new services or ways of interacting with traditional services and strengthening the relationship between a bank and its customers. One such example is the use of gestures in AR. The concept of enhanced banking is the introduction of digital channels and architectural solutions that will be implemented on mobile devices and the Web. The expanded bank offers new payment options. The enhanced banking system will provide a fully immersive experience through artificial intelligence-related scenarios, such as data visualization and visual recognition, and it will integrate into the new digital life (Mergaliyev et al. 2021).

An impressive feature of augmented banking is that it helps customers make decisions based on relevant real-world information. For example, if a customer considers selling a car, he/she can use the smartphone through the camera to view important information such as price, car loan, current value, and value change in recent years. Based on this information, the client knows the best time to sell a car. Augmented banking offers banks a great opportunity to build strong relationships with customers seeking information and assistance in fulfilling their aspirations. Augmented banking provides an immersive experience similar to what customers have in their daily lives. New mechanisms can support customers in making timely decisions. By providing a more vivid financial experience, augmented banking can translate value into unique recommendations for customers (Zheng et al. 2019).

Inclusive finance

The United Nations proposed inclusive finance in 2005, emphasizing that by improving financial infrastructure, financial services would be extended to underdeveloped regions and low-income groups at an affordable cost, providing them with accessible, convenient, and fast financial services. The main purpose of inclusive finance is to provide

savings, financial, and credit services to all social groups with the ability to repay, especially small and micro enterprises, low-income groups in urban and rural areas, migrant workers, the poor, the disabled, and the elderly. The goal is to establish a fair, efficient, and comprehensive peer-to-peer service financial system (Sarma and Pais 2011).

Owing to these defects, traditional finance has limitations in providing financial services for small- and medium-sized enterprises (SMEs), rural areas, and underdeveloped areas. This shows both the financial structure and financial discrimination. Factors such as geography, income, age, and sex affect traditional financing. The participating institutions are more diversified and the products are more abundant. Inclusive financing can effectively allocate financial resources and help to realize social equity. However, there are many obstacles and challenges in the process of realizing inclusive finance. Examples include (1) insufficient financial service capacity, (2) high costs that can make it difficult to benefit from large demand, (3) information barriers and data isolation, and (4) regional gaps in the financial inclusion system (Ozili 2021).

These problems can be solved based on the features of blockchain. First, it can reduce inclusive financial transaction costs and improve transaction efficiency. Second, blockchain can expand the coverage of financial services and provide financial benefits. Third, blockchain can be used to build a new credit system and control potential financial risks. The blockchain and financial fields are constantly exploring integration and development, with blockchain technology promoting the development of inclusive finance, especially the development and progress of digital inclusive finance (Siddik and Kabiraj 2020).

Quantum finance

Quantum mechanics originates from physics and has great computing power when dealing with financial calculation problems, such as uncertainty estimation, investment optimization, and portfolio simulation. The financial market is dynamic and changeable, and the acceleration of quantum computing reflects real market fluctuations and meets customer needs. The existing financial system has certain limitations, and a quantum system can solve these issues as a good supplement to traditional systems. Well-designed products and services are delivered to customers in a secure network with fewer time constraints. Quantum computing leverages the principles of quantum mechanics to perform complex calculations at speeds that are unimaginable using classical computers. In finance, quantum computing holds promise for solving complicated mathematical problems, optimizing portfolios, and enhancing encryption methods (Egger et al. 2020).

However, traditional computing approaches have reached certain limits. When solving a problem on a parallel computer, according to the algorithm, one finds that on a certain number of processors, convergence can be achieved from the perspective of speedup. Quantum computing makes algorithms exponentially faster. Two examples are financial crisis forecasting, especially in today's volatile markets, and understanding how the financial system behaves at the micro and macroeconomic levels when certain things happen (Orús et al. 2019).

Quantum computing is still in its infancy. We want to build a platform and a community to discuss issues in quantum computing and finance. What are relevant future cases? Why is high performance computing still used? We believe that wherever

accelerated computing is required in the future, there will be a hybrid of quantum and high-performance computing (Zoufal et al. 2019).

Conclusion

The conventional financial paradigm faces a multitude of challenges that cannot be resolved through existing mechanisms. Finance presents compelling prospects for integrating technologies to accommodate the multifaceted and evolving exigencies inherent in business operations. Specifically, the AI, machine learning, blockchain, AR/VR, and quantum mechanics domains gained substantial attention and assume critical roles in the advancement of the financial sector. These emerging technologies collectively contribute to creating a financial ecosystem that is more precise, optimized, secure, and agile, thereby meeting the evolving demands of the industry and ensuring better outcomes for stakeholders. This study explores the significance and operational attributes of embedding these emerging technologies in financial systems.

Fintech can be enhanced by incorporating qualitative evidence or quantitative analyses to illustrate real-world applications or experiments in the convergence of finance and technology. Moreover, effectively balancing the merits and demerits of these novel technologies constitutes an important consideration that profoundly influences the efficacy of their assimilation into financial operations and transactions. Compared with the prevailing financial architecture, a well-architected infrastructure rooted in emerging technologies has the potential to generate heightened value and is poised to establish itself as a central source of competitive advantage.

Abbreviations

AI	Artificial intelligence
AR	augmented reality/VR (virtual reality)
ATM	Automated Teller Machine
EBSCO	EBSCO Publishing
WoS	Web of Science

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Declarations

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